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When Your Threat Model *Lives With You*

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Conventional threat models do not anticipate attackers who possess intimate knowledge of, and access to, victims.

> **The IPV tech-abuse field has been making this point for years.** This study is what that adversary looks like at scale.

Eight thousand four hundred twenty-eight apps.
One adversary who lives with the victim.

A different shape of adversary.

TRADITIONAL

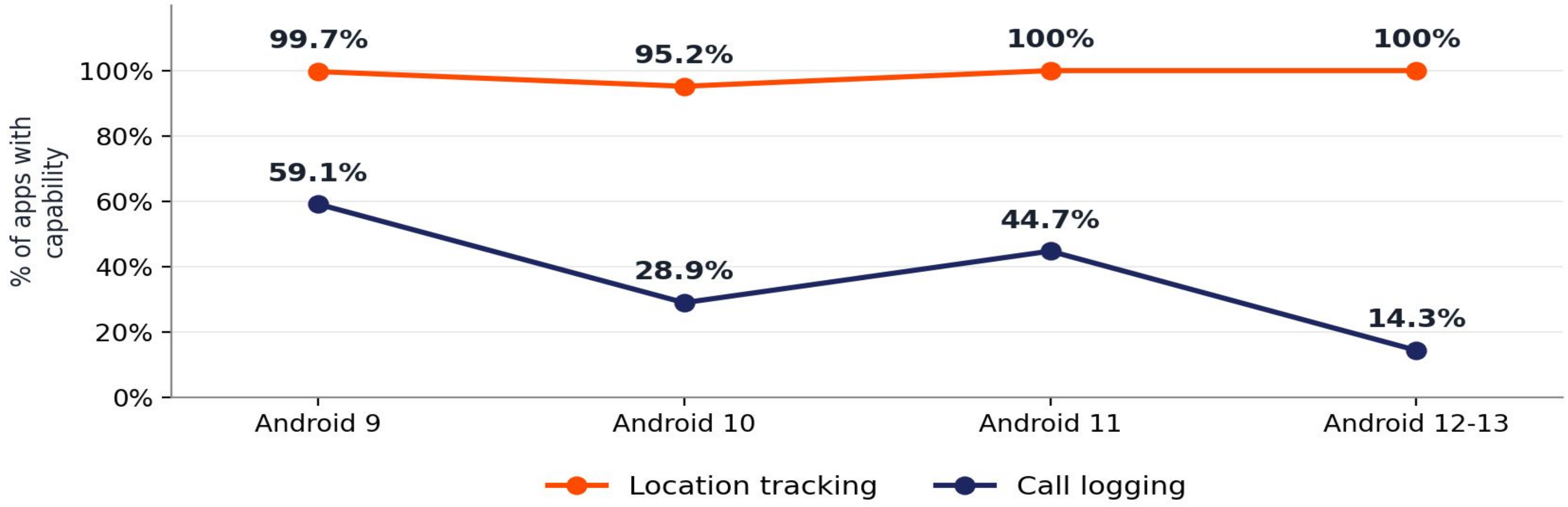
- **Remote attacker**
- No physical access
- Exploits vulnerabilities
- Loses access when detected

STALKERWARE (IPV)

- **Partner with physical access**
- Installs the app themselves
- Retaliates when caught

Google did the right things.

Since 2018, Android tightened permissions each version.



Capabilities persisted. Adaptation, not elimination.

Some defenses cannot be hardened further.

ACCESSIBILITY SERVICES

APIs designed for users with visual or motor accessibility needs. Stalkerware uses them to read messaging apps.

BACKWARD COMPATIBILITY

Apps target older Android versions. Android preserves old APIs to maintain backward compatibility.

ROOT + DEVICE ADMIN

44% expects rooted devices. 32% requests device-admin. Privileges someone with physical access can grant.

Stalkerware operates inside structural limits.

Some apps exist only to teach the adversary.

No surveillance code. Request no permissions. Not flagged by filters.

765

apps - 9.08% of our corpus

Walking through how to sideload the actual app
Disabling Play Protect
Granting permissions covertly

Detection misses them by design.
They contain nothing to detect.

The disclosure layer is where detection hasn't looked yet.

- The 765 apps contain instructions, not malware — and instructions are detectable.
- What apps claim to do (UI strings) often diverges from what they do (code).
- That gap — claims vs. behavior — is what disclosure analysis closes.

Look where the abuser already does.

The device that delivered the threat could be gift.

That is the kind of threat model this is!

PAPER

Malvika Jadhav, Wenxuan Bao, Vincent Bindschaedler.

[*Thwart Me If You Can. arXiv:2508.02454*]

Thank you.

refugetechsafety.org

National DA Helpline : 0808 2000 247

Coalition Against Stalkerware:

stopstalkerware.org

Tech Abuse Clinics : techabuseclinics.org

Scan or visit: https://malvikajadhav.github.io/london-2026/

Email: jadhav.m@ufl.edu

Citations: scale of stalkerware and its IPV context.

31,000+ unique individuals affected across 175 countries (2023).

Kaspersky. The State of Stalkerware in 2023. Public report, 2024.

70% of women victims of cyberstalking also experienced physical or sexual IPV.

WWP EN, Coalition Against Stalkerware launch materials, Nov 2019 — citing studies.

Stalking and IPV co-occur at population scale — tracked nationally.

CDC NISVS (ongoing); DOJ Office of Justice Programs — Stalking Victimization reports.

Coalition Against Stalkerware founded Nov 2019 to bridge IT security and DV advocacy.

stopstalkerware.org — 10 founding partners include EFF, Kaspersky, NNEDV, Malwarebytes, NortonLifeLock.

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How we built the corpus, and what we did with it.

8,428 Android stalkerware samples through Dec 2022; 8,421 decompiled successfully.

Source: Coalition Against Stalkerware threat list. JADX + JEB Pro for decompilation.

Two-phase static analysis: permissions/APIs, then user-visible strings.

Section 3.2. Capability taxonomy adapted from Parsons et al. 2019 (nine capabilities).

Validated against FlowDroid taint analysis on a 50-app random sample.

Section 4.4. Our method matched or exceeded FlowDroid in coverage.

Dynamic analysis via ADB on emulators matching each sample's targetSDKVersion.

Section 3.2. All unique APKs targeting Android 9+ installed and observed.

Detailed capability percentages from the corpus.

Location 90.66% · SMS 70.44% · Call logging 53.77% · Ambient audio 25.65%.
Screenshots 43.18% · Accessibility screen reading 7.12%.

Section 4.2, Table 5. Percentages of 8,421 samples possessing each capability.

44.12% expect rooted devices; 31.75% request device-administrator privileges.

Section 6.2. Both elevations are grantable by a user with physical access.

765 apps (9.08%) request no permissions; their sole function is installation guidance.

Section 6.1. Pass Play Protect because they contain no malicious behaviors.

UI strings appear in 103 distinct languages across the corpus; 17.19% non-English primarily.

Section 6.5. Indicates distribution channels developed for non-English-speaking abusers.

What this study does not cover — and what does.

Dual-use apps (Find My–type tracking) are out of scope for this study.
Almansoori et al. A global survey of dual-use Android apps used in IPS. PoPETs, 2022.

We do not address detection accuracy of antivirus products on stalkerware.
Han et al. Towards Stalkerware Detection with Precise Warnings. ACSAC, 2021.

Our corpus reflects the state of mitigation up to and including Android 13.
Android 14+ accessibility-service restrictions for sideloaded apps post-date the corpus.

Findings establish lower bounds — obfuscation and anti-reverse-engineering impede analysis.
Section 8. Static + dynamic + taint analyses used to mitigate undercounting.

Why we say “without consent” — and Google's actions.

“Without adequate notice or consent, no persistent notification.”

Google Play Developer Program Policy — stalkerware definition, effective Oct 1, 2020.

Stalkerware is distributed outside the Play Store via sideloading.

Section 2; corpus collected from Coalition Against Stalkerware threat list, not Play Store.

Google's platform timeline: runtime permissions (6), scoped storage (10), privacy indicators (12).

Android Developer Documentation — Android version release notes, 2015 onward.

October 2020: stalkerware banned from Play Store; ad policy followed.

Google Play Developer Program Policy update, Sept 16, 2020; effective Oct 1, 2020.